25900301 **MAHATMA GANDHI UNIVERSITY, KOTTAYAM**

MGU – UGP (HONOURS) REGULAR EXAMINATION OCTOBER 2025

THIRD SEMESTER

DSC – MG3DSCCOM201- QUANTITATIVE TECHNIQUES FOR BUSINESS

(2024 ADMISSION ONWARDS)

Maximum Marks: 70 **Duration: 2 Hours**

Scheme for Valuation

Part A

Very Short Answer Questions. **Answer any 12 questions**.

Each question carries **1 mark**.

1. Descriptive Statistics: Summarising and describing the features of the data in a study using various measures. (like central tendency, dispersion etc.)
2. Relative measures of dispersion (Coefficient of Variation or co-efficient of mean deviation etc.)- Award marks for the usage relative measure or any specific relative measure.
3. Harmonic Mean
4. To study the variability in a distribution (income or wealth inequality)
5. **27.52**. (Pearson Skewness Coefficient (Jp) = (Mean – Mode)/SD = 29.6- (6.5 x 0.32)
6. **25** (Mode = 3Median – 2Mean)
7. Mean deviation =**2** (Mean will be 5 and sum of absolute deviations is 8.)
8. The **determinant** must be non-zero.
9. Primary data is collected by researcher himself, so it is original/first in hand/ genuine/raw (consider any other suitable answers) in nature.
10. Platykurtic
11. Stable or consistent
12. One
13. A - B = $[-6 \quad -5]$
14. The body of a table refers to the main part containing the actual data or numerical values that are being presented or analysed.

Part B

Short Answer Questions. **Answer any 4 questions**.

Each question carries **7 marks**.

15. Statistical averages must be used with caution because ***they can hide crucial variations within the data, be easily skewed by extreme values, and provide a misleading representation of the entire situation if the data is not normally distributed***.
    - Averages are designed for quantitative data. Calculating an "average" for qualitative characteristics (like honesty or intelligence) or categorical data (like the average number of wheels on different types of vehicles) can result in a meaningless or absurd figure
    - Averages only show a midpoint and not the entire range or spread of results
    - Statistical conclusions are only as good as the data they are based on.

    (*Consider any relevant point*- 4 points + explanation)
16. Calculation of Mean Deviation from Mean (Deviation/Step deviation method etc. can also be used)

Marks	X (mid value)	F	fx	|d|(x-25.5)	f |d|
0 – 10	5	10	50	20.5	205
10 – 20	15	25	375	10.5	262.5
20 – 30	25	30	750	0.5	15
30 – 40	35	20	700	9.5	190
40 – 50	45	15	675	19.5	292.5
		N =100	Σ fx =2550		Σ f |d| =965

Mean = Σ fx /n = 2550/100 = **25.5 (3 marks)**
Mean Deviation from Mean = Σ f |d|/n = 965/100 = **9.65 (3 marks)**
Coefficient of MD = MD/Mean = 9.65/25.5 = **0.378 (1 mark)**
(Step mark shall be awarded if the procedure is correctly followed.

17. Pearsonian Coefficient of Skewness

Marks: 35 36 37 38 39 40 42 45

Since there's no clear mode, use Pearson's second formula:
Sk (J) = $\frac{3\ (\text{Mean}-\text{Median})}{\text{SD}}$

- Mean $\bar{x}$ = (35+36+37+38+39+40+42+45)/8 = **39**
- Median = (38+39)/2= 38.5
- Standard deviation:

$$\frac{\sqrt{\sum(x-\bar{x})^2}}{N}$$

Deviations(d) from 39: (−4, −3, −2, −1, 0, 1, 3, 6))

- Squares: 16, 9, 4, 1, 0, 1, 9, 36
- squares sum $\Sigma d^2 = 76$
  σ = √76/8 = √9.5
  =3.0822
- Sk = 3(39−38.5)/3.0822
  = 1.5/3.082
  = 0.487

The distribution is positively skewed. (low or slight positive skewness)
(Calculation of mean and median= 2 marks; Standard deviation = 2 marks- Identification of formula – 1 mark; Calculation of skewness- 1 mark and comment- 1 mark)
Note- Instead, mode can be calculated using empirical relation and different formula can be applied

18. Profit Margin Matrix P = $\begin{bmatrix}15 & 20\\12 & 18\end{bmatrix}$ - (The same can be expressed as 0.15, 0.20 etc as the figures are in percentages) **(3 marks)**

For matrix:

P = $\begin{bmatrix}a & b\\c & d\end{bmatrix}$, The adjoint is: adj(P) = $\begin{bmatrix}d & -b\\-c & a\end{bmatrix}$

Adjoint of Matrix P

adj(P) = $\begin{bmatrix}18 & -20\\-12 & 15\end{bmatrix}$ **(4 marks)**

(Instead, adjoint can be calculated by calculating the minors, preparing the cofactor matrix and finding its transpose)

19. Limitations of Questionnaire *(Any relevant 6 points- 3 marks plus explanation – 4 marks)*

* Low response rate
* Lack of personal contact
* Misinterpretation of questions
* Incomplete responses
* Limited to literate respondents
* Time-consuming for complex topics
* Language barriers
* No control over who answers
* Inflexibility
* No opportunity to probe
* Difficulty in capturing emotions
* Pre-determined questions only
* Potential for dishonest answers
* Costly for large samples

20. $\bar{x}_{12} = \frac{N_1\bar{x}_1 + N_2\bar{x}_2}{N_1 + N_2}$

Worker category	Ram Nagar			Shyam Nagar		
	Number	Wage per hour (₹)	Total wage per hour ($N_1\bar{x}_1$ & $N_2\bar{x}_2$)	Number	Wage per hour (₹)	Total wage per hour ($N_1\bar{x}_1$ & $N_2\bar{x}_2$)
Skilled	150	180	27000	350	175	61250
Unskilled	850	130	110500	650	125	81250
Total	1000		**137500**	1000		**142500**

Average hourly wage for Ram Nagar = 137500/1000 = ₹ 137.50
Average hourly wage for Shyam Nagar = 142500/1000 = ₹ 142.50

(Identification of formula/approach- 2 marks- and 2.5 marks each for the communities)

Part C

Short Answer Questions. **Answer any 2 questions**.
Each question carries 15 **marks**.

21. Calculation of Median

No. of days absent (X)	No. of students (f)	Cf
0-5	29	29
5-10	195	224
10-15	241	465
15-20	117	582
20-25	52	634
25-30	10	644
30-35	6	650
35-40	3	653
40-45	2	655
	N = 655	

(Calculation of frequencies and cumulative frequencies= 5 marks)

m = size of $(N/2)^{th}$ item = size of $(655/2)^{th}$ item = **327.5^{th} item**
Median class = **10 – 15 (Median class determination – 2 marks)**

Median = $L1 + \frac{L2 - L1}{F}(m-c)$ *(Can be presented in different forms)* – **1 mark**

L1 = 10; L2 = 15; f = 241; m = 327.5; c = 224

$= 10 + \frac{15 - 10}{241} \times (327.5-224)$

$= 10 + \frac{5}{241} \times (103.50)$

= 10 + 2.1473 = 12.1473 = **12.15 (Application and answer = 7 marks)**

22. $SkB = \frac{(Q3+Q1 - 2\,Median)}{Q3-Q1}$

Class Interval	**Frequency (f)**	**Cumulative Frequency (CF)**
0–10	12	12
10–20	15	27
20–30	17	44
30–40	25	69
40–50	8	77
50–60	5	82
60–70	2	84
Total (N)	**84**	

Step 1: Locate Quartiles

N = 84 Q1 at N/4 = 84/4 = 21th value
Median at N/2 = 84/2 = 84/2 = 42nd value
Q3 at 3N/4=3(84)/4=252/4 = 63rd value

(Location of classes- 2 marks)

Step 2: Find Each Quartile Using Formula

Quartiles $= L + \frac{(kN4 - CF)}{f} * c$

Where k is the quartile position, first, second or third quartile

(i) Q1 (21st item)

Falls in 10–20
L=10, CF prev =12, f=15, C =10 L=10
Q1= 16

(ii) Median (42nd item)

Falls in 20–30 (since CF up to 10–20 = 27 < 42 ≤ 44).
L=20, CF prev =27, f=17, C=10 L=20
=20+8.82=**28.82**

(iii) Q3 (63rd item)

Falls in 30–40 (since CF up to 20–30 = 44 < 63 ≤ 69).
L=30, CF prev =44, f=25, h=10 L=30
=30+7.6= **37.6**

(9 marks for calculations of Q1, M and Q3)

Step 3: Bowley's Coefficient

$$SkB = \frac{(Q3+Q1-2\ Median)}{Q3-Q1}$$

$$= \frac{37.6+16-2(28.82)}{37.6-16}$$

$$= -4.04 / 21.6 = \mathbf{-0.187}$$

Bowley's Coefficient of Skewness = –0.19 (approx.)

(Formula and calculation of skewness – 3 marks)

The distribution is slightly negatively skewed (longer left tail). **(Comment – 1 mark)**

23. To find the total daily production cost for each shift, multiply the production matrix P by the cost matrix C. (Daily Production Cost Matrix = P × C)

Calculating Each Shift's Total Cost:

Shift 1 (Row 1):

(120 × 250) + (150 × 180) + (180 × 320) = 30,000 + 27,000 + 57,600

= **₹114,600**

Shift 2 (Row 2):

Cost = (100 × 250) + (140 × 180) + (160 × 320) = 25,000 + 25,200 + 51,200

= **₹101,400**

Shift 3 (Row 3):

(110 × 250) + (130 × 180) + (170 × 320) = 27,500 + 23,400 + 54,400

= **₹105,300**

Shift 4 (Row 4):

(90 × 250) + (120 × 180) + (150 × 320) = 22,500 + 21,600 + 48,000

= **₹92,100**

$$\text{Daily Cost Matrix} = \begin{bmatrix} 114600 \\ 101400 \\ 105300 \\ 92100 \end{bmatrix} \begin{bmatrix} Shift\ 1 \\ Shift\ 2 \\ Shift\ 3 \\ Shift\ 4 \end{bmatrix}$$

(Application of matrix multiplication – **2** marks.
2.5 marks each for cost of each shift = **10** marks total)

a) Which shift has the highest total production cost?
Shift 1 has the highest total production cost of ₹114,600 **(1.5 marks)**

b) Which shift has the lowest total production cost?
Shift 4 has the lowest total production cost of ₹92,100 **(1.5 marks)**